E-commerce Order Fulfillment & Customer Behavior Analysis

1. Objective

The objective of this project is to analyze order lifecycle, delivery performance, and customer behavior in an e-commerce dataset. The goal is to identify key operational strengths and areas for improvement, particularly in delivery efficiency and customer retention.

2. Dataset Overview

The dataset contains information on customer orders, order status, delivery timelines, and product details. It primarily includes completed transactions, making it suitable for analyzing fulfillment performance and customer purchasing behavior.

3. Approach

The analysis began with a funnel overview to understand user progression. Due to minimal drop-offs in the dataset, the focus shifted to delivery performance, customer segmentation, and revenue analysis. SQL was used for data extraction and transformation, including joins across multiple tables to analyze customer behavior and revenue patterns. Power BI was used for visualization.

4. Key Insights

- ~97% of orders are successfully delivered, indicating strong fulfillment performance.
- Average delivery time is approximately 12.5 days.
- Only ~3% of customers are repeat users, indicating low customer retention.
- A significant portion of revenue is concentrated in a few product categories, indicating key business drivers.
- A large proportion of customers are one-time buyers, highlighting an opportunity to improve customer lifetime value.

5. Recommendations

- Implement customer retention strategies such as loyalty programs and targeted promotions.
- Optimize delivery timelines to improve customer satisfaction.
- Focus on high-performing product categories to maximize revenue.
- Analyze customer feedback to identify potential service improvements.

6. Tools Used

- SQL (MySQL) — data extraction and transformation
- Power BI — interactive dashboards and visualizations
- Excel / CSV — data handling and preprocessing

7. Conclusion

The analysis highlights strong operational efficiency in order fulfillment but reveals a key gap in customer retention. Addressing retention challenges can significantly improve long-term business growth and customer lifetime value.